Near-term Quantum Computing — lecture notes

Stefano Polla

2026-10-15

Table of contents

Near-term Quantum Computing 3

1 Platforms, gates and measurement 4
- 1.1 Noisy qubits and fault tolerance . . . 4
- 1.2 Physical qubits . . . 5
  - 1.2.1 What a qubit needs . . . 5
  - 1.2.2 Superconducting qubits . . . 6
  - 1.2.3 Trapped ions . . . 7
  - 1.2.4 Other platforms . . . 8
- 1.3 Native gates and connectivity . . . 9
  - 1.3.1 Pauli rotations . . . 9
  - 1.3.2 How gates are driven . . . 11
  - 1.3.3 Native gate sets . . . 11
  - 1.3.4 Virtual Z gates . . . 12
  - 1.3.5 Connectivity . . . 13
- 1.4 Where noise comes from . . . 15
- 1.5 Measurement . . . 16
  - 1.5.1 How readout works . . . 16
  - 1.5.2 Shots and expectation values . . . 17
  - 1.5.3 Timescales . . . 17
  - 1.5.4 Shot noise . . . 18
- 1.6 Summary . . . 19
- 1.7 Further reading . . . 19

References . . . 19

Appendices 21

A Glossary 21

Near-term Quantum Computing

Lecture notes for *Near-term Quantum Computing* (UvA, 2026-2027), one chapter per lecture. A PDF version can be downloaded from the menu.

The notes assume basic quantum mechanics and the circuit model: Dirac notation, Pauli matrices, density matrices, expectation values and their statistical error.

Version: 2026-10-15 (draft).

1 Platforms, gates and measurement

On paper, a quantum algorithm is a circuit, a sequence of gates on ideal qubits followed by measurements. A real quantum computer is a physical device, and its physics decides which operations it can perform, between which qubits, how fast and how reliably. This chapter is about that physical side and what it means for the circuits we can run. By the end of it, you should be able to explain what a physical system needs to serve as a qubit, and how superconducting circuits and trapped ions meet those needs. You should be able to explain why devices have different native gates, why rotations about the z-axis are usually free, and what limited connectivity costs. And you should be able to estimate how many shots an expectation value needs, and how long they take on each platform.

1.1 Noisy qubits and fault tolerance

Every physical qubit is imperfect. Each gate has a small chance of going wrong, the stored quantum state degrades even when nothing is done to it, and measurements sometimes return the wrong result. These errors add up. If each gate fails with probability 0.1%, a circuit of a thousand gates runs without any error only about 37% of the time, since $0.999^{1000} \approx e^{-1}$. Noise therefore limits the size of the circuits that a device can run usefully.

Preskill called the devices of this era *NISQ* devices, for *noisy intermediate-scale quantum* (Preskill 2018). *Intermediate scale* means tens to thousands of qubits, enough that many of their circuits are hard to simulate on a classical computer, but far from the millions of qubits that the best-known quantum algorithms need. *Noisy* means that their errors are not corrected.

Correcting errors is possible in principle. *Quantum error correction* stores each qubit of information redundantly in many *physical qubits*, which together form one *logical qubit*. Repeated checks on the physical qubits reveal errors without disturbing the stored information, so that the errors can be undone. This only helps if the physical qubits are good enough. Below a certain error rate per operation, the *threshold*, larger codes give lower logical error rates, as low as a computation requires, and the computer is called *fault tolerant*.

Fault tolerance is expensive. Useful algorithms are estimated to need of order a thousand physical qubits for every logical qubit, and around a million physical qubits in total (Fowler et al. 2012; Gidney 2025). Experiments are now crossing the threshold: in 2024, a superconducting device ran an error-correcting code whose logical error rate halved each time the code was

made larger (Google Quantum AI and Collaborators 2025). Such experiments use a whole device to protect a single logical qubit, however, and useful computations still run on physical qubits. We come back to the stage in between, *early fault tolerance*, at the end of the course.

This course works with physical qubits. We run circuits on them, estimate how much their noise affects the results, and use methods that reduce its effect without full error correction. One question will come back throughout: when a device produces a number, what would it take to believe it?

1.2 Physical qubits

The device physics in this section is descriptive, and you are not expected to derive it. What matters later are its consequences for gates, connectivity and speed. Terms we leave unexplained are in the glossary; on the web version, hover over an underlined term to see its definition.

1.2.1 What a qubit needs

A qubit is a quantum system with two states, $|0\rangle$ and $|1\rangle$, that we can control. Almost every quantum system has more than two states, so most qubits are the two lowest energy levels of a larger system, such as an atom or an electrical circuit. The other levels are still there, and a qubit can end up in them by accident.

For a two-level system to work as a qubit, we must be able to

- prepare it in a known state, usually $|0\rangle$;
- apply gates to it, both rotations of a single qubit and an interaction between two qubits, which together form a universal set of gates;
- measure it;
- keep its quantum state long enough, which means that its *coherence time* must be long compared with the duration of a gate.

Together with the requirement that the system can be scaled up to many qubits, these are the *DiVincenzo criteria* (DiVincenzo 2000).

The requirements pull in opposite directions. A qubit keeps its state longest when it is well isolated from its surroundings, since every interaction with the environment disturbs it. Yet to apply gates and to measure, we have to interact with it, strongly and quickly. Every platform is a compromise between isolation and control, and this compromise explains many of the numbers in this chapter.

1.2.2 Superconducting qubits

Quantum mechanics links energy and frequency. An energy difference E between two levels corresponds to a frequency $f = E/h$, where h is Planck's constant, and radiation at that frequency drives transitions between the two levels, while radiation at other frequencies hardly affects them. This selectivity is called *resonance*. It is how individual qubits are addressed: to rotate a qubit, we send it a pulse at its own frequency.

A superconducting qubit is a small electrical circuit on a chip. Once cooled, its metal becomes superconducting. Current then flows without resistance, so no energy is lost as heat, and the circuit behaves as a quantum system with discrete energy levels; we treat this part as a black box. A simple circuit of this kind has evenly spaced levels, like a mass on a spring (Figure 1.1, left). That makes it useless as a qubit, because a pulse at the frequency f_{01} of the $|0\rangle \to |1\rangle$ transition is just as resonant with $|1\rangle \to |2\rangle$, $|2\rangle \to |3\rangle$ and so on. A circuit element called a *Josephson junction* makes the spacing uneven. In the most common design, the *transmon*, the $|1\rangle \to |2\rangle$ transition is a few percent lower in frequency than the qubit transition (Figure 1.1, right), so a pulse at f_{01} drives only the qubit.

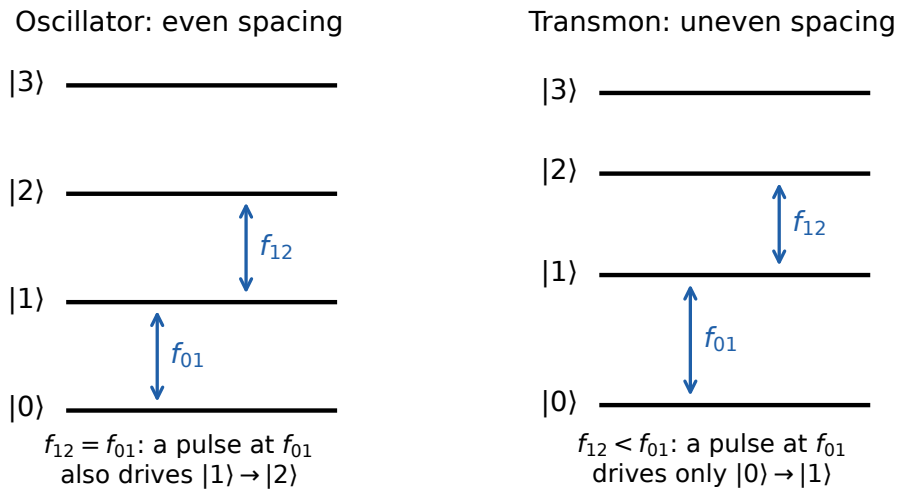


Figure 1.1: Energy levels of an oscillator and of a transmon. With even spacing, every transition has the same frequency, so the two lowest levels cannot be driven on their own. The transmon's uneven spacing makes the qubit transition unique. Not to scale: in a real transmon, f_{12} is only a few percent below f_{01}.

Transmon frequencies are typically 4-6 GHz, in the microwave range. At room temperature, the thermal energy is about a thousand times larger than hf, so heat alone would excite the qubit at random. Superconducting chips therefore sit in a *dilution refrigerator* at about 10 mK, a hundredth of a degree above absolute zero (Exercise 1.1), and cables bring the microwave pulses down to them from electronics at room temperature.

Because the qubits are fabricated, no two of them come out identical. Each has its own frequency, its own coherence time and its own error rates, which the provider's calibration data list qubit by qubit. The couplings between qubits are built into the chip, which fixes which pairs can interact. Superconducting gates are fast, taking tens to hundreds of nanoseconds, but the quantum state of a qubit typically survives only for tens to hundreds of microseconds.

Exercise 1.1 (Why so cold?)**.** In thermal equilibrium at temperature T, the excited and ground states of a qubit with frequency f are populated in the ratio $p_1/p_0 = e^{-hf/k_BT}$ (the Boltzmann factor), where k_B is Boltzmann's constant. Use $h/k_B \approx 0.048$ K/GHz.

(a) Compute p_1/p_0 for a 5 GHz qubit at room temperature (300 K), at 1 K and at 10 mK.

(b) Below which temperature is p_1/p_0 smaller than 10^{-3}?

1.2.3 Trapped ions

A trapped-ion qubit is a single *ion*, an atom with one electron removed, typically of ytterbium, barium or calcium. Two of its internal energy levels form the qubit. All ions of an element are exactly the same, so, unlike fabricated circuits, the qubits of an ion-trap computer are identical by nature.

The ions are held in vacuum by electric fields in an *ion trap*, lined up in a chain a few micrometres apart, and laser beams slow them down until they are almost at rest (*laser cooling*). Laser beams, or in some designs microwaves, also drive the gates; a beam focused on one ion rotates that ion's qubit. The qubit levels can be chosen so that stray magnetic fields and other disturbances hardly affect them, and trapped-ion qubits keep their quantum state for a second or more, thousands of times longer than superconducting qubits.

The ions are charged and repel each other, so the chain behaves like a row of balls joined by springs (Figure 1.2). Push one ion, and the whole chain starts to vibrate. Like the energy of an atom, the energy of these vibrations comes in discrete units, called *phonons*. Every ion takes part in the same vibrations, so they act as a shared channel, the *phonon bus*, through which any ion can interact with any other. The two-qubit gates are built on it.

This comes at the cost of speed. Gates that use the phonon bus take from tens of microseconds to about a millisecond, roughly a thousand times longer than superconducting gates, and they become slower and harder to control as the chain grows. To scale up, some devices, such as Quantinuum's, divide the trap into zones and move ions between them, bringing the two ions of each gate together (*QCCD*).

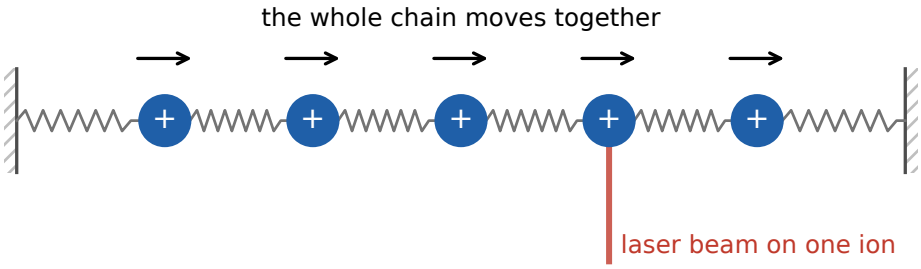


Figure 1.2: A trapped-ion chain as balls joined by springs. The ions repel each other, so they move together; shown here is the vibration in which the whole chain moves as one. A laser beam focused on a single ion drives that ion's qubit.

1.2.4 Other platforms

Several other platforms are developing quickly, following the same recipe of a two-level system, a way to drive it, an interaction between qubits and a readout. In *neutral-atom* devices, uncharged atoms are held, and can be moved, by tightly focused laser beams (*optical tweezers*), and they interact when lasers excite them to very large *Rydberg states*; arrays of hundreds to thousands of atoms have been built (Henriet et al. 2020). *Photonic* devices carry qubits on particles of light, which hardly interact with their surroundings but, for the same reason, hardly with each other either, so their entangling operations succeed only some of the time (Slussarenko and Pryde 2019). *Spin qubits* use the *spin* of single electrons trapped in tiny semiconductor structures (*quantum dots*); they can be made with the techniques of the chip industry, but current devices have at most a few tens of qubits (Burkard et al. 2023). Table 1.1 compares the five platforms.

Table 1.1: Platforms for gate-based quantum computing, compared.

Platform	Qubit	Two qubits interact through	Strengths	Main limitations
Superconducting	energy levels of a circuit on a chip	couplings built into the chip	fast gates; mature fabrication	short coherence; fixed connectivity; qubits not identical
Trapped ions	energy levels of an ion	shared vibrations of the chain	identical qubits; long coherence; any pair can interact	slow gates; long chains are hard to control

Platform	Qubit	Two qubits interact through	Strengths	Main limitations
Neutral atoms	energy levels of an atom	excitation to large Rydberg states	many qubits; atoms can be rearranged	slow; atoms get lost and must be replaced
Photonics	a photon	interference and detection	little decoherence; photons travel far	entangling operations often fail; photons get lost
Spin qubits	spin of an electron	neighbouring electrons	very small; chip-industry fabrication	few qubits so far; qubits not identical

1.3 Native gates and connectivity

A circuit as written in an SDK is not what runs on a device. The hardware implements a small set of *native gates*, each one a calibrated control pulse, and two-qubit gates only between pairs of qubits that are physically coupled. Every other gate must be compiled into these.

1.3.1 Pauli rotations

Recall the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

They are Hermitian and square to the identity, $X^2 = Y^2 = Z^2 = I$. Any two different Pauli matrices anticommute, for example $XZ = -ZX$, and the product of two of them is the third, up to a factor: $XY = iZ$, $YZ = iX$ and $ZX = iY$. On several qubits, we combine them with tensor products.

Definition 1.1 (Pauli string)**.** A *Pauli string* on n qubits is a tensor product $P = P_1 \otimes P_2 \otimes \cdots \otimes P_n$ with each $P_j \in \{I, X, Y, Z\}$. We often drop the tensor product signs and write, for example, ZZ or XIY.

A Pauli string is again Hermitian, and it squares to the identity, because the square of a tensor product is the tensor product of the squares: $P^2 = P_1^2 \otimes \cdots \otimes P_n^2 = I$. This one property gives a closed formula for the exponential of P. Expand the exponential as a power series. Even

powers of P are the identity, $P^{2m} = I$, and odd powers are $P^{2m+1} = P$, so the series splits into two:

$$e^{-i\theta P/2} = \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m)!} \left(\frac{\theta}{2}\right)^{2m} I \; - \; i \sum_{m=0}^{\infty} \frac{(-1)^m}{(2m+1)!} \left(\frac{\theta}{2}\right)^{2m+1} P.$$

These are the power series of the cosine and the sine, so

$$e^{-i\theta P/2} = \cos\frac{\theta}{2}\, I - i \sin\frac{\theta}{2}\, P. \tag{1.1}$$

The derivation used only $P^2 = I$, so Equation 1.1 holds for any operator with that property, not only for Pauli strings.

Applied to single Pauli matrices, Equation 1.1 gives the familiar rotations of a qubit about the axes of the Bloch sphere, for example

$$R_X(\theta) = e^{-i\theta X/2} = \begin{pmatrix} \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} \\ -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}, \qquad R_Z(\theta) = e^{-i\theta Z/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}. \tag{1.2}$$

Rotations about two different axes are enough to build any single-qubit gate. Up to a global phase, every single-qubit unitary can be written as $R_Z(\alpha)\, R_Y(\beta)\, R_Z(\gamma)$ for some angles (Nielsen and Chuang 2010, sec. 4.2).

Applied to the two-qubit Pauli string $Z \otimes Z$, Equation 1.1 gives

$$R_{ZZ}(\theta) = e^{-i\theta\, Z\otimes Z/2} = \mathrm{diag}\big(e^{-i\theta/2}, e^{i\theta/2}, e^{i\theta/2}, e^{-i\theta/2}\big) \tag{1.3}$$

in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. R_{ZZ} applies a phase that depends only on whether the two bits are equal or different, their parity. That is why it can be built from two CNOTs and a single R_Z (Figure 1.3). The first CNOT writes the parity of the two bits into the second qubit, R_Z applies the phase, and the second CNOT undoes the first.

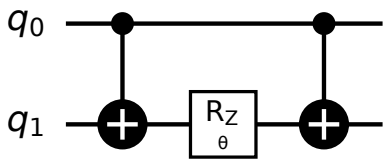


Figure 1.3: $R_{ZZ}(\theta)$ built from two CNOTs and one $R_Z(\theta)$.

Equation 1.1 will come back several times in this course, in circuits for time evolution, which are built from Pauli rotations, and in the parameter-shift rule, which gives exact gradients of variational circuits.

1.3.2 How gates are driven

The native gates of a platform are not an arbitrary choice. They are what its physical interactions do easily.

A single-qubit gate on a superconducting qubit is a short microwave pulse, a few tens of nanoseconds long, at the qubit's frequency. Since each qubit has its own frequency, the pulse leaves the other qubits almost untouched. The strength and duration of the pulse set the rotation angle θ. Its phase, that is, the timing of the microwave oscillation relative to a reference clock, sets the axis of rotation, which lies in the xy-plane of the Bloch sphere at an angle φ from the x-axis. Laser or microwave pulses on trapped ions work in the same way. On both platforms, the natural single-qubit gate is therefore a rotation by any angle about any axis in the xy-plane, which IQM calls PRX:

$$\mathrm{PRX}(\theta, \varphi) = \exp\left[-i\frac{\theta}{2}\left(\cos\varphi\, X + \sin\varphi\, Y\right)\right]. \tag{1.4}$$

The operator $\cos\varphi\, X + \sin\varphi\, Y$ squares to the identity, so Equation 1.1 applies to it too. For $\varphi = 0$ and $\varphi = \pi/2$, PRX reduces to $R_X(\theta)$ and $R_Y(\theta)$. Rotations about the z-axis are missing from this family, and Section 1.3.4 shows that they need no pulse at all.

During a two-qubit gate, the two qubits should interact, and at all other times they should not. On most current superconducting devices, a tunable element between two neighbouring qubits switches their interaction on for a short time, and holding it on for the right time produces a controlled-Z (CZ) gate. Older IBM devices instead have couplings that are always on, and drive one qubit at the frequency of its neighbour (*cross-resonance*). This produces a gate called ECR, which is equivalent to a CNOT up to single-qubit rotations.

On trapped ions, laser beams on two ions couple their qubit states to the shared vibrations of the chain. Through the phonon bus, the two qubits interact. At the end of the gate the vibration returns to its starting state, so that it carries away no information about the qubits. This *Mølmer-Sørensen gate* is a rotation generated by $X \otimes X$, of the form of Equation 1.1, with an angle set by the duration and strength of the laser pulses. It is available at any angle, not only as a maximally entangling gate. Combined with single-qubit rotations, it gives $R_{ZZ}(\theta)$ at any angle, the native two-qubit gate of current trapped-ion devices.

1.3.3 Native gate sets

Table 1.2 lists the native gates of some current devices. Names and details change with each hardware generation, but the pattern does not: a continuous family of single-qubit rotations and one entangling gate, sometimes two. Together they are universal, so any circuit can be compiled into them, at the cost of extra gates. Several entries in the table are vendor names for xy-plane rotations. IBM's $\sqrt{X}$ is $R_X(\pi/2)$ up to a global phase, Google's PhasedXZ is an

xy-plane rotation followed by a Z rotation, and IonQ's GPi and GPi2 are rotations by π and $\pi/2$ about an axis in the xy-plane.

Table 1.2: Native gates of some current devices, as of 2026. All of them also offer virtual Z rotations (Section 1.3.4).

Device family	Single-qubit	Two-qubit
IQM (superconducting)	$\mathrm{PRX}(\theta, \varphi)$	CZ
IBM (superconducting)	$\sqrt{X}$, X	CZ (newer devices) or ECR (older devices)
Google (superconducting)	PhasedXZ	CZ
Quantinuum (trapped ions)	$R_{xy}(\theta, \varphi)$, the same as PRX	$R_{ZZ}(\theta)$, any angle
IonQ (trapped ions)	GPi, GPi2	$ZZ(\theta)$, any angle

1.3.4 Virtual Z gates

PRX contains no rotations about the z-axis, yet circuits are full of them. The phase gates S and T are Z rotations, and so are many parametrised gates in algorithms. On most devices, superconducting and trapped-ion alike, Z rotations are implemented without any pulse, as *virtual Z gates*. The trick follows from Pauli algebra.

Conjugation by a Z rotation rotates X and Y within the xy-plane:

$$R_Z(\alpha)\, X\, R_Z(\alpha)^\dagger = \cos\alpha\, X + \sin\alpha\, Y, \qquad R_Z(\alpha)\, Y R_Z(\alpha)^\dagger = \cos\alpha\, Y - \sin\alpha\, X$$

(Exercise 1.2). By the angle-addition formulas, it therefore maps $\cos\varphi\, X + \sin\varphi\, Y$ to $\cos(\varphi + \alpha)\, X + \sin(\varphi + \alpha)\, Y$. Conjugation also commutes with the exponential, $Ue^A U^\dagger = e^{UAU^\dagger}$ for any unitary U, so conjugating a PRX pulse by a Z rotation shifts its phase:

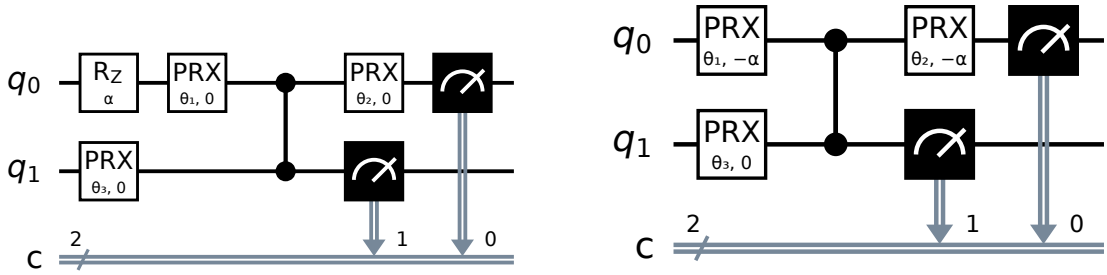
$$R_Z(\alpha)\, \mathrm{PRX}(\theta, \varphi)\, R_Z(\alpha)^\dagger = \mathrm{PRX}(\theta, \varphi + \alpha). \tag{1.5}$$

Replacing α by $-\alpha$ and multiplying by $R_Z(\alpha)$ on the left gives

$$\mathrm{PRX}(\theta, \varphi)\, R_Z(\alpha) = R_Z(\alpha)\, \mathrm{PRX}(\theta, \varphi - \alpha). \tag{1.6}$$

Operators act from right to left, so on the left-hand side $R_Z(\alpha)$ acts first and the pulse second. On the right-hand side, the pulse acts first, with its phase shifted by $-\alpha$. A Z rotation can therefore be moved later in a circuit, past a pulse, by changing the pulse's phase.

Now take a circuit with a Z rotation on one qubit (Figure 1.4a), and move the Z rotation forward in time. Each PRX it passes gets its phase shifted by $-\alpha$. CZ is diagonal, like R_Z, so the two commute and the Z rotation passes through unchanged. At the end, the Z rotation sits just before a measurement in the computational basis. There it only changes the phases of $|0\rangle$



(a) With a Z rotation on q_0.

(b) The Z rotation absorbed into the phases of the later pulses on q_0.

Figure 1.4: Two circuits that give the same measurement statistics.

and $|1\rangle$, not the probabilities of measuring them, so it can be dropped. The result (Figure 1.4b) contains no Z rotation at all.

In hardware, the phase of each pulse, microwave or laser, is measured against a reference clock in the control electronics, and a virtual Z gate shifts that reference for all later pulses on the qubit. It is an instruction to the electronics, carried out instantly and exactly, so Z rotations take no time and add no error. When we estimate the error of a circuit, only the other gates count.

The same trick makes every single-qubit gate cost a single pulse. Write the gate as $R_Z(\alpha)\,R_Y(\beta)\,R_Z(\gamma)$, note that $R_Y(\beta) = \mathrm{PRX}(\beta, \pi/2)$, and move $R_Z(\gamma)$ past the pulse with Equation 1.6:

$$R_Z(\alpha)\,R_Y(\beta)\,R_Z(\gamma) = R_Z(\alpha+\gamma)\,\mathrm{PRX}(\beta, \pi/2-\gamma).$$

That is one pulse, followed by a Z rotation that is virtual.

Exercise 1.2 (Virtual Z gates)**.**

(a) Show that $R_Z(\alpha)\,X\,R_Z(\alpha)^\dagger = \cos\alpha\,X + \sin\alpha\,Y$ and $R_Z(\alpha)\,Y R_Z(\alpha)^\dagger = \cos\alpha\,Y - \sin\alpha\,X$. Hint: write $R_Z(\alpha)$ with Equation 1.1, and use the products of Pauli matrices.

(b) Write the Hadamard gate, up to a global phase, as a single PRX pulse followed by a Z rotation.

1.3.5 Connectivity

A native two-qubit gate is available only between qubits that are physically coupled. The *coupling graph* of a device records which pairs these are. Its vertices are the qubits, and an edge joins two qubits if a native two-qubit gate can act on them (Figure 1.5).

On a superconducting chip, the coupling graph is fixed when the chip is made. In a square lattice, used by Google, IQM and IBM's newest devices, each qubit has up to four neighbours. IBM's heavy-hex lattice, used on most of its current devices, gives each qubit only two or three, which reduces unwanted interactions between qubits but lengthens the routes between distant ones. Some IQM devices use a star layout, in which every qubit couples to a central resonator, a circuit element that can take over the state of one qubit and pass it on to another; in Figure 1.5 it is drawn as an extra vertex.

In a trapped-ion chain, every ion couples to the same phonon bus, so every pair can interact and the coupling graph is complete. Devices that move ions between zones achieve the same by bringing any two ions together.

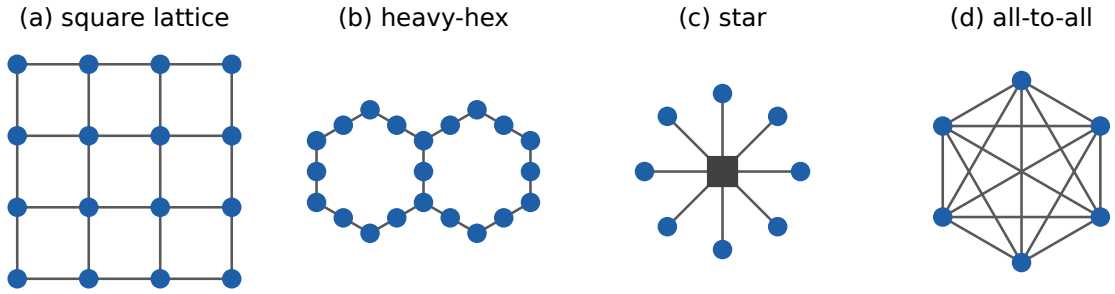


Figure 1.5: Coupling graphs: (a) square lattice; (b) heavy-hex lattice; (c) star, with every qubit coupled to a central resonator (square); (d) all-to-all, as in a trapped-ion chain.

What if a circuit needs a two-qubit gate between qubits that are not neighbours? Their states must first be brought together with *SWAP* gates, which exchange the states of two neighbouring qubits, and each SWAP costs three CNOTs (Figure 1.6). A gate between distant qubits can therefore cost many extra two-qubit gates (Exercise 1.3). Two-qubit gates are the main source of error on current devices (Section 1.4), so connectivity directly limits which circuits a device can run reliably.

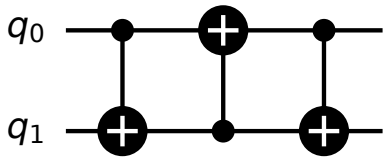


Figure 1.6: SWAP as three CNOTs with alternating control.

In practice, nobody inserts SWAPs by hand. *Transpilation* maps a circuit onto the native gates and coupling graph of a device. It chooses which physical qubit plays the role of each qubit in

the circuit, inserts SWAPs where needed, and simplifies the result. We return to its cost in later chapters.

Exercise 1.3 (The cost of distance)**.**

(a) Check Figure 1.6 on the four computational basis states.

(b) On a line of n qubits, how many CNOTs does a CNOT between the two end qubits cost if the qubits may stay in their new positions afterwards? And if they must be returned to their original positions?

1.4 Where noise comes from

Every wire or laser beam used to control a qubit is also a way in for disturbances. This section describes in words what can go wrong, and the next chapter gives each effect its mathematical description.

The most important effect is that an excited qubit loses its energy. Left alone, $|1\rangle$ decays to $|0\rangle$, a process called *relaxation*. In superconducting qubits, the energy is absorbed by tiny defects in the materials of the chip. The qubit levels of trapped ions hardly decay on their own, but the lasers used for gates occasionally knock an ion out of its qubit state.

A qubit can also lose its phase without losing energy. A superposition of $|0\rangle$ and $|1\rangle$ carries a relative phase between its two parts, and the qubit's frequency sets how fast this phase turns. If the frequency fluctuates, for example because of fluctuating magnetic fields or electric charges near the qubit, the phase wanders, and the superposition gradually becomes a random mixture of $|0\rangle$ and $|1\rangle$. This is called *dephasing*. The time over which a qubit relaxes is called T_1, and the time over which it dephases is called T_2. These are the coherence times quoted for every device, and the next chapter defines them precisely.

A third kind of error is a gate that rotates by the wrong amount, because a pulse is slightly too long, too short or off axis, or because a laser's intensity fluctuates. Unlike relaxation and dephasing, such *coherent errors* are systematic: the same gate makes the same mistake every time. Table 1.3 lists these effects together with the remaining ones.

Table 1.3: What can go wrong in a quantum device.

What happens	Example	Name in the next chapter
$\|1\rangle$ decays to $\|0\rangle$	energy absorbed by defects in the chip	relaxation (T_1)
the relative phase of a superposition wanders	fluctuating magnetic fields	dephasing (T_2)

What happens	Example	Name in the next chapter
a gate rotates by the wrong amount	a pulse slightly too long	coherent error
the qubit leaves the qubit levels	a pulse pushes a transmon into its third level (Figure 1.1)	*leakage*
an operation on one qubit disturbs another	couplings that are never fully off; a laser beam that partly hits the next ion	*crosstalk*
the readout is wrong	0 reported as 1, or the reverse	readout error
the device drifts	properties that change over hours	(*recalibration*)

The last row explains why providers recalibrate their devices regularly and publish calibration data, the error rates of each qubit and gate at the last calibration. These data are a snapshot, and a few hours later the device may behave differently.

Providers summarise the quality of each operation by a *fidelity*. For now, read it as one minus the probability that the operation goes wrong; the next chapter gives the precise definition. Typical values on current devices are 99.9-99.99% for single-qubit gates, 99-99.9% for two-qubit gates and 96-99.9% for readout, so two-qubit gates and readout are the weakest links.

1.5 Measurement

1.5.1 How readout works

On superconducting devices, each qubit is coupled to a small resonant circuit, its readout resonator, whose frequency shifts slightly depending on whether the qubit is in $|0\rangle$ or $|1\rangle$. To measure, the electronics send a weak microwave pulse to the resonator and decide between 0 and 1 from the signal that comes back (*dispersive readout*). This takes about a microsecond. On trapped ions, a laser beam is tuned so that an ion in one qubit state scatters light and appears bright, while an ion in the other state stays dark. A camera or photon detectors count the light from each ion, and this fluorescence readout takes from a hundred microseconds to a millisecond.

On both platforms, a measurement returns one bit per qubit, in the computational basis. It projects each qubit onto $|0\rangle$ or $|1\rangle$, and measuring in another basis means first rotating that basis onto the computational basis.

1.5.2 Shots and expectation values

A single run of a circuit, from preparing the qubits to measuring them, is called a *shot*. It returns one bit string, drawn at random from the distribution given by the Born rule, so to learn anything quantitative we repeat the circuit many times.

Most quantities in this course are expectation values $\langle P \rangle$ of Pauli strings. To estimate one, we rotate each qubit so that its factor of P becomes Z (nothing for Z, a Hadamard for X, and $S^\dagger$ followed by a Hadamard for Y), and measure. A Z measurement with outcome bit b corresponds to the eigenvalue $(-1)^b$ of Z, so the outcome of P is the product of these signs over the qubits on which P acts:

$$s = (-1)^{\sum_{j:\, P_j \neq I} b_j} = \pm 1.$$

For $P = ZZ$, for example, $s = +1$ if the two bits agree and $s = -1$ if they differ. The average of s over N shots,

$$\bar{s} = \frac{1}{N} \sum_{k=1}^{N} s_k, \tag{1.7}$$

is our estimate of $\langle P \rangle$. A rule like this, which turns data into an estimate, is called an *estimator*.

1.5.3 Timescales

Table 1.4 collects typical timescales of current devices. They vary between devices and improve every year, so read them as orders of magnitude.

Table 1.4: Typical timescales of current devices, as orders of magnitude.

Time per	Superconducting	Trapped ions
single-qubit gate	20-60 ns	10-150 µs
two-qubit gate	20-700 ns	50 µs-1 ms
readout	0.5-2.5 µs	0.1-1 ms
coherence (T_2)	20-350 µs	about 1 s or more
shot (reset, circuit, readout)	0.3-4 ms	10 ms to several seconds
switching to a new circuit	much longer than a shot, plus any queue	the same

The coherence time divided by the two-qubit gate time is between about a thousand and ten thousand on both platforms, although their speeds differ by a factor of about a thousand. Coherence alone would allow circuits of thousands of gates in sequence. In practice, gate errors of 0.1-1% per two-qubit gate limit useful circuits much earlier.

Where the time goes differs between the platforms. On superconducting devices, the gates are fast, and most of a shot is spent measuring the qubits and resetting them to $|0\rangle$; devices run hundreds to thousands of shots per second. Switching to a different circuit costs much more, since the circuit must be compiled and loaded into the control electronics, and on a shared cloud device it may wait in a queue. Many shots of one circuit are cheap, many different circuits are not. On trapped ions, the gates themselves are slow, and each shot also includes cooling the ions and, on some devices, moving them between zones, which takes most of the time. There, the number of shots is the main cost.

1.5.4 Shot noise

How many shots does an estimate need? Each shot gives $s = +1$ or $s = -1$ at random, and the expectation value of s is $\langle P \rangle$. The sample mean $\bar{s}$ of Equation 1.7 is therefore correct on average, or *unbiased*. Its random error comes from the spread of single shots. Since $s^2 = 1$ for every shot (this is $P^2 = I$ again), the variance of a single shot is

$$\mathrm{Var}(s) = \langle s^2 \rangle - \langle s \rangle^2 = 1 - \langle P \rangle^2.$$

The mean of N independent samples has $1/N$ times the variance of a single sample (for a refresher on this and other basic statistics, see Blitzstein and Hwang (2019)). The typical size of the random error of $\bar{s}$, its *standard error*, is therefore

$$\sigma_{\bar{s}} = \sqrt{\frac{1 - \langle P \rangle^2}{N}}. \tag{1.8}$$

In practice $\langle P \rangle$ is unknown, and we evaluate Equation 1.8 with $\bar{s}$ in its place.

The error is largest when $\langle P \rangle = 0$, where every shot is a coin flip, and vanishes when $\langle P \rangle = \pm 1$, where every shot gives the same result. It decreases only as $1/\sqrt{N}$, so each extra digit of precision costs a hundred times more shots. To estimate an expectation value near zero with a standard error of 0.01, for example, we need $N = 1/0.01^2 = 10^4$ shots. By Table 1.4, that takes seconds to a minute on a superconducting device, and from a quarter of an hour to a few hours on an ion trap.

Shot noise is not the only error in an estimated expectation value. Noise in the device also shifts $\langle P \rangle$ itself, usually towards zero, and more shots cannot remove that shift; they only estimate the shifted value more precisely. The next chapter separates these two kinds of error. Tutorial 1 practises Equation 1.7 and Equation 1.8 in code: open the T1 notebook in Colab.

Exercise 1.4 (A shot budget)**.**

(a) How many shots give a standard error of 0.01 when $\langle P \rangle \approx 0.9$? Compare with $\langle P \rangle \approx 0$.

(b) You need ten expectation values, each from its own circuit, each with a standard error of at most 0.005. Assume the worst case, $\langle P \rangle = 0$. Using Table 1.4, estimate the total time on a superconducting device and on an ion trap. Which part of the time dominates on each?

1.6 Summary

- Noise limits the size of circuits on current devices. Error correction will lift the limit, at a cost of about a thousand physical qubits per logical qubit.
- A qubit is two controllable levels of a larger system that keep their quantum state long enough. Isolation and control pull in opposite directions.
- Superconducting qubits are fast, each has its own frequency, and their couplings are fixed. Trapped ions are identical, long-lived and fully connected through their shared vibrations, but slow.
- Native gates are what the physics does easily: rotations about axes in the xy-plane and one entangling gate. Z rotations are virtual and free, and two-qubit gates between distant qubits cost SWAPs.
- A measurement gives one bit per qubit. The standard error of a Pauli expectation value is $\sqrt{(1-\langle P\rangle^2)/N}$, so precision costs shots, and shots cost time.

1.7 Further reading

For superconducting qubits, Krantz et al. (2019) is a thorough guide written for engineers; for trapped ions, see Bruzewicz et al. (2019). Henriet et al. (2020), Slussarenko and Pryde (2019) and Burkard et al. (2023) review neutral atoms, photonics and spin qubits. Preskill (2018) introduced the NISQ era, and Nielsen and Chuang (2010, chap. 4) covers the circuit model.

References

Blitzstein, Joseph K., and Jessica Hwang. 2019. *Introduction to Probability*. 2nd ed. Chapman; Hall/CRC. https://probabilitybook.net.

Bruzewicz, Colin D., John Chiaverini, Robert McConnell, and Jeremy M. Sage. 2019. "Trapped-Ion Quantum Computing: Progress and Challenges." *Applied Physics Reviews* 6 (2): 021314. https://doi.org/10.1063/1.5088164.

Burkard, Guido, Thaddeus D. Ladd, Andrew Pan, John M. Nichol, and Jason R. Petta. 2023. "Semiconductor Spin Qubits." *Reviews of Modern Physics* 95 (2): 025003. https://doi.org/10.1103/RevModPhys.95.025003.

DiVincenzo, David P. 2000. "The Physical Implementation of Quantum Computation." *Fortschritte Der Physik* 48 (9–11): 771–83. https://doi.org/10.1002/1521-3978(200009)48:9/11%3C771::AID-PROP771%3E3.0.CO;2-E.

Fowler, Austin G., Matteo Mariantoni, John M. Martinis, and Andrew N. Cleland. 2012. "Surface Codes: Towards Practical Large-Scale Quantum Computation." *Physical Review A* 86 (3): 032324. https://doi.org/10.1103/PhysRevA.86.032324.

Gidney, Craig. 2025. *How to Factor 2048 Bit RSA Integers with Less Than a Million Noisy Qubits*. https://arxiv.org/abs/2505.15917.

Google Quantum AI and Collaborators. 2025. "Quantum Error Correction Below the Surface Code Threshold." *Nature* 638: 920–26. https://doi.org/10.1038/s41586-024-08449-y.

Henriet, Loïc, Lucas Beguin, Adrien Signoles, et al. 2020. "Quantum Computing with Neutral Atoms." *Quantum* 4: 327. https://doi.org/10.22331/q-2020-09-21-327.

Krantz, Philip, Morten Kjaergaard, Fei Yan, Terry P. Orlando, Simon Gustavsson, and William D. Oliver. 2019. "A Quantum Engineer's Guide to Superconducting Qubits." *Applied Physics Reviews* 6 (2): 021318. https://doi.org/10.1063/1.5089550.

Nielsen, Michael A., and Isaac L. Chuang. 2010. *Quantum Computation and Quantum Information*. 10th anniversary. Cambridge University Press.

Preskill, John. 2018. "Quantum Computing in the NISQ Era and Beyond." *Quantum* 2: 79. https://doi.org/10.22331/q-2018-08-06-79.

Slussarenko, Sergei, and Geoff J. Pryde. 2019. "Photonic Quantum Information Processing: A Concise Review." *Applied Physics Reviews* 6 (4): 041303. https://doi.org/10.1063/1.5115814.

A Glossary

calibration Measuring and adjusting the control pulses of a device so that its gates do what they should. Providers publish calibration data: the error rates of each qubit and gate at the last calibration.

coherence time How long a qubit keeps its quantum state, in particular a superposition, before noise from its surroundings destroys it.

coherent error A systematic error in which a gate applies the wrong unitary, for example a rotation by a slightly wrong angle, in the same way every time.

coupling graph Graph whose vertices are qubits and whose edges are the pairs on which a native two-qubit gate is available.

cross-resonance A two-qubit gate mechanism for superconducting qubits with fixed couplings: one qubit is driven at the frequency of its neighbour.

crosstalk An unwanted effect of an operation on one qubit on other qubits, for example through couplings that are never fully switched off.

dephasing Loss of the relative phase of a superposition, for example because the qubit frequency fluctuates. Its characteristic time is called T2.

dilution refrigerator A refrigerator that reaches about 10 millikelvin, cold enough that heat no longer excites superconducting qubits.

dispersive readout Readout of a superconducting qubit through a resonator whose frequency depends slightly on the qubit state. A weak microwave probe tells the two states apart.

DiVincenzo criteria A checklist for building a quantum computer: scalable, well-characterised qubits; preparation of a known state; long coherence; a universal set of gates; measurement of individual qubits.

estimator A rule that turns measured data into an estimate of a quantity, for example the sample mean of the outcomes as an estimate of an expectation value.

fidelity A number between 0 and 1 measuring how close an operation or state is to the intended one. Informally, one minus the probability of an error. Defined precisely in the chapter on noise.

ion An atom with one or more electrons removed, so that it carries an electric charge and can be held by electric fields.

ion trap A device that holds ions in vacuum with electric fields, typically lined up in a chain a few micrometres apart.

Josephson junction A thin insulating barrier between two superconductors. In a circuit it makes the energy levels unevenly spaced, which is what turns the circuit into a usable qubit.

laser cooling Slowing the motion of atoms or ions with laser light until they are almost at rest.

leakage An error in which a qubit leaves its two qubit levels, for example when a transmon is pushed into its third level.

logical qubit A qubit of information encoded in many physical qubits by an error-correcting code. Its error rate can be much lower than that of the physical qubits.

Mølmer-Sørensen gate The standard two-qubit gate of trapped ions. Laser beams couple two ions to the shared vibrations of the chain, producing an entangling rotation of any chosen angle.

native gate A gate the hardware implements directly with a calibrated control pulse. Every other gate must be compiled into native gates.

NISQ Noisy intermediate-scale quantum: devices with tens to thousands of qubits whose errors are not corrected, so noise limits the size of the circuits they can run.

optical tweezers Tightly focused laser beams that hold single neutral atoms in place and can move them.

Pauli string A tensor product of single-qubit Pauli matrices (I, X, Y or Z), one for each qubit, such as ZZ or XIY. Hermitian, and squares to the identity.

phonon A quantum of vibration. In an ion trap, phonons are the quanta of the vibrations that the whole chain of ions shares.

phonon bus The shared vibrations of a chain of trapped ions, used as a common channel through which any two ions can interact in a two-qubit gate.

physical qubit A qubit as implemented in hardware, for example one transmon or one ion, with all its imperfections.

QCCD Quantum charge-coupled device: a trapped-ion architecture in which ions are moved between zones of the trap, so that any two ions can be brought together for a gate.

quantum dot A tiny region in a semiconductor that can trap single electrons.

quantum error correction Encoding quantum information redundantly in many physical qubits, so that errors can be detected and corrected without disturbing the encoded information.

relaxation Loss of energy by a qubit: the excited state decays to the ground state. Its characteristic time is called T1.

resonance Radiation drives a transition between two energy levels only when its frequency matches their energy difference divided by Planck's constant. This is how individual qubits are addressed.

Rydberg state A state of an atom in which one electron is excited very far from the nucleus. Atoms in Rydberg states interact strongly over micrometre distances.

shot One run of a circuit: prepare the qubits, apply the gates, measure. Returns one bit string.

spin An intrinsic two-level property of an electron (up or down), which can serve as a qubit.

SWAP A two-qubit gate that exchanges the states of two qubits. It costs three CNOTs.

threshold The physical error rate per operation below which error correction helps: below it, larger codes give lower logical error rates.

transmon The most common superconducting qubit: a small circuit with a Josephson junction, whose two lowest energy levels form the qubit.

transpilation Compilation of a circuit into the native gates and coupling graph of a device, including inserting SWAPs and simplifying.

virtual Z gate A Z rotation implemented by shifting the phase of all later drive pulses instead of applying a pulse. It takes no time and adds no error.